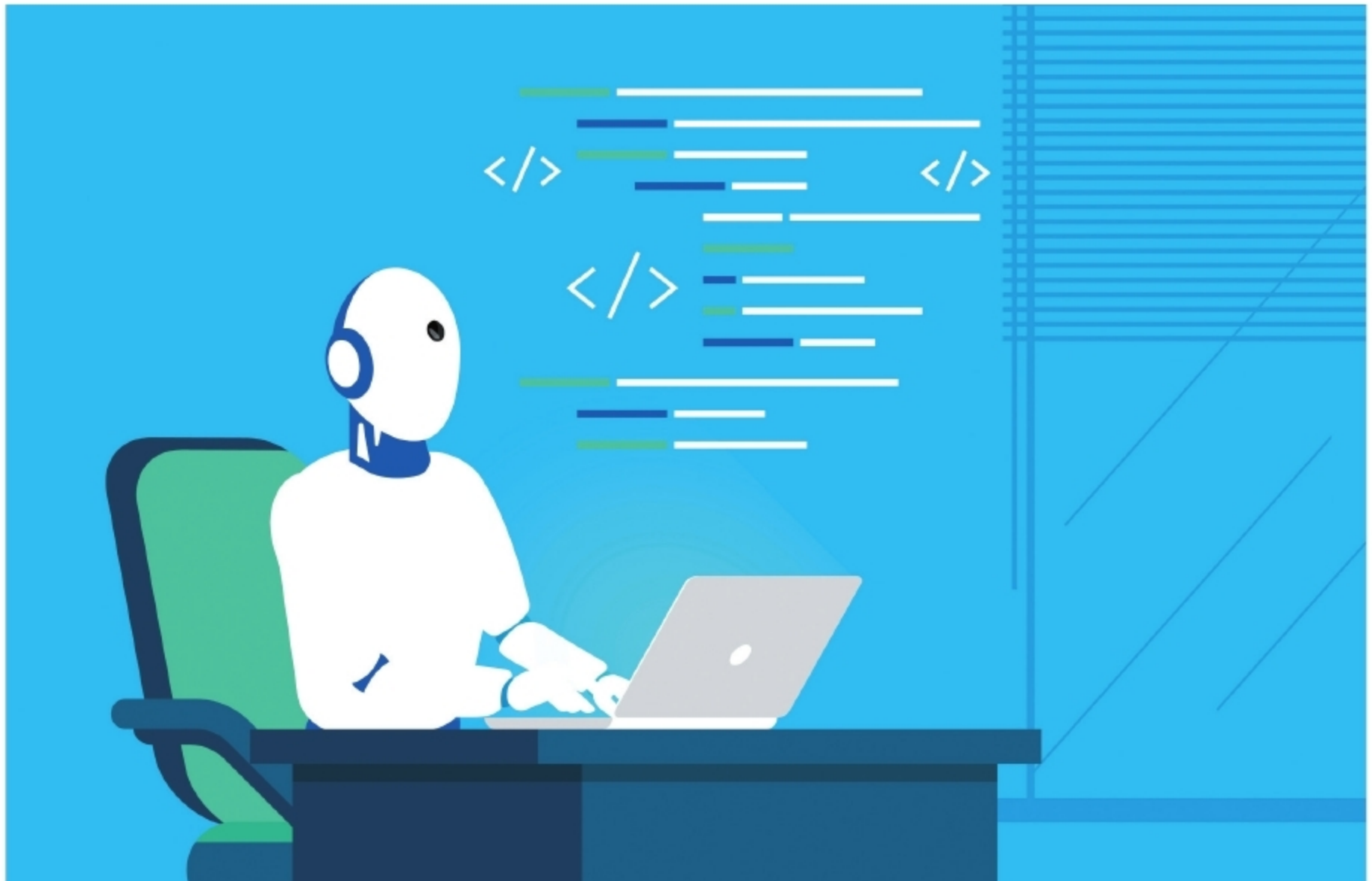


The Capabilities of **Tensor Virtual Machine, an Open Deep Learning Compiler Stack**



The Tensor Virtual Machine stack began as a research project at the SAMPL (System, Architecture, Machine learning and Programming Language) group of the Paul G. Allen School of Computer Science & Engineering, at the University of Washington in the US. This project is now driven by an open source community, and involves multiple industry and academic institutions.

Tensor Virtual Machine or TVM is an open deep learning compiler stack to compile various deep learning models from different frameworks to the CPU, GPU or specialised accelerators. TVM supports model compilation from a wide range of frontends like TensorFlow, Onnx, Keras, Mxnet, Darknet, CoreML and Caffe2. TVM-compiled modules can be deployed on backends like LLVM (JavaScript or WASM, AMD GPU, ARM or X86), NVIDIA GPU (CUDA), OpenCL and Metal. TVM also supports runtime bindings for programming languages like JavaScript, Java, Python, C++ and Golang. With a wide range of frontend, backend and runtime bindings, this deep learning

compiler enables developers to integrate and deploy deep learning models from any framework to any hardware, via any programming language.

The TVM architecture

TVM provides a two-level optimisation mechanism, as shown in Figure 1. The first level of optimisation happens at the graph level after a model is imported. This optimisation provides graph level fusion, layout transformation and memory management. Later optimisation happens at the tensor level – at the code-generation layer and is based on the paper <https://arxiv.org/abs/1802.04799>.